

DESIGN AND SIMULATION OF A WEIGHTED MATCHING SYSTEM FOR VOLUNTARY ACADEMIC SUPPORT.

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INTRODUCTION

Vulnerable communities in Bogotá face a critical shortage of human resources for personalized academic support, significantly increasing the risk of school dropouts. Although university students are willing to volunteer, the coordination process is typically manual, disorganized, and difficult to measure. The primary challenge is to act as a technological mediator, transforming this inefficient process into a traceable and optimizable digital system.

GOAL

How can an automated digital system optimize the connection between volunteers and educational projects to maximize academic impact? The expected product is a platform based on a three-tier architecture that utilizes a weighted matching algorithm to ensure precise tutor-student pairings.

EXPERIMENTS

Two computational simulations were conducted to validate the system’s design:

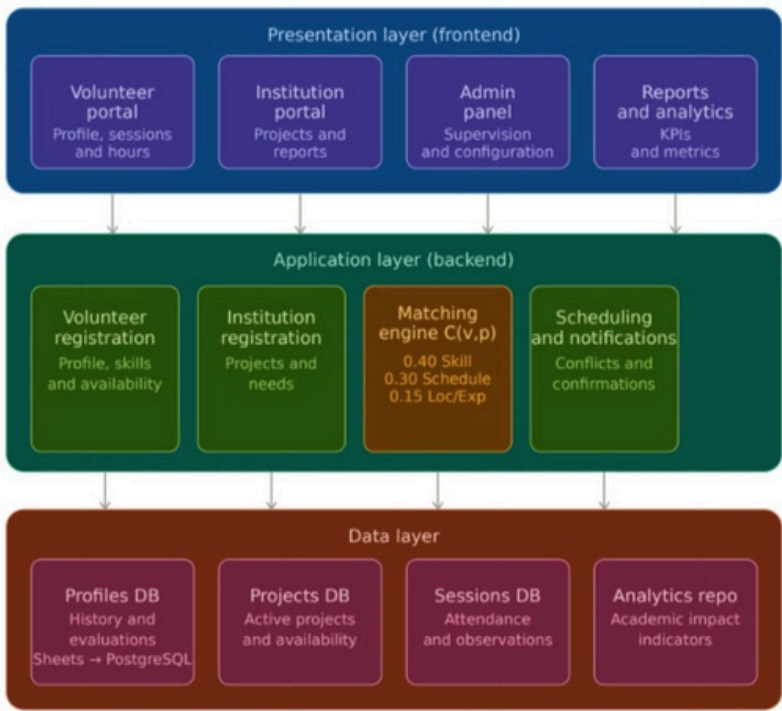
Discrete-Event Simulation (DES): Modeled the 12-week assignment pipeline under three scenarios: Baseline (50 volunteers), Optimistic (80), and Stress (25).

Cellular Automata (CA): Modeled engagement propagation in a 30 x 30 university grid to observe emergent network effects and the risk of "burnout contagion."

Scenario	Volunters	Projects	Rationale
Baseline	50	20	Current operational reality from survey data.
Stress	25	20	Performance evaluation under critical volunteer shortages.

PROPOSED SOLUTION

The system is structured into three levels—Presentation (Frontend), Application (Backend Logic), and Data (Databases)—to ensure scalability and modularity. The core engine is the Matching Algorithm, which calculates compatibility based on four dimensions: Skills (40%), Schedule (30%), Location (15%), and Experience (15%).



RESULTS

The algorithm achieved mean compatibility scores of 0.860 to 0.887, comfortably exceeding the 0.80 design target. The platform reached 100% project coverage in all scenarios with at least 25 active volunteers. Additionally, the system accelerated engagement by 18% through social network emergence. A "Quality Paradox" was identified: match quality is highest under resource scarcity (Stress scenario) because the algorithm concentrates on the most compatible pairings.

CONCLUTIONS

The platform efficiently automates tutor linking, achieving response times of less than 60 seconds and high assignment accuracy. The system is viable for 80+ concurrent users, but requires a minimum floor of 30 volunteers for stable coverage. Identifying "burnout contagion" through simulation adds a vital new layer to risk management for long-term project sustainability.